

SOC104 — Malware Detection & Response Investigation

LetsDefend SOC Analyst Simulation | Role: Tier 1 SOC Analyst | MITRE ATT&CK: T1204 – User Execution

Alert	Final Verdict	Severity	Playbook Score
SOC104 – Malware Detected	True Positive	High	100%

1. Summary

A malicious executable (**Invoice.exe**) was flagged on host **AdamPRD** and allowed to execute. Investigation confirmed the file was malware, identified an associated command-and-control (C2) IP address using open-source threat intelligence, checked internal logs for evidence of C2 communication, and contained the affected endpoint. The case was closed as a **True Positive**.

This report documents the investigation exactly as it was performed: the alert that triggered it, the evidence reviewed at each step, the tools used, and the reasoning behind each decision.

2. Alert Details

Field	Value
Event ID	36
Event Time	2020-12-01T10:23:12+03:00
Rule	SOC104 – Malware Detected
Severity	High
Alert Type	Malware
File Name	Invoice.exe
File Hash (MD5)	f83fb9ce6a83da58b20685c1d7e1e546
File Size	473.00 KB
Device Action	Allowed
Source Address	10.15.15.18
Source Hostname	AdamPRD

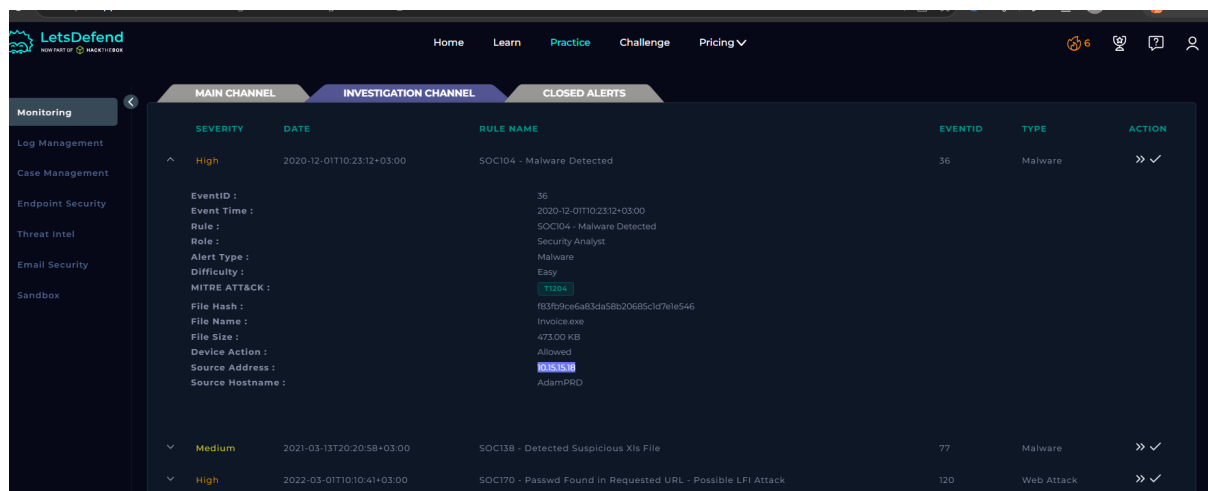


Figure 1 — Alert detail showing the malware event fields.

A file named **Invoice.exe** is a classic social-engineering lure pattern, and the alert shows the device **allowed** rather than blocked the file — meaning it likely executed on the host before detection.

3. Investigation Steps

Step 1 — Triage the alert

Reviewed the alert fields above and noted the host, file hash, and the fact that the device action was “Allowed” (not blocked), which raised the priority of the investigation.

Step 2 — Analyze the file / identify infrastructure

Ran the associated indicators through open-source malware analysis tooling (VirusTotal). This surfaced a related IP address, **92.63.8.47**, flagged by threat intelligence as C2 infrastructure associated with Maze ransomware activity.

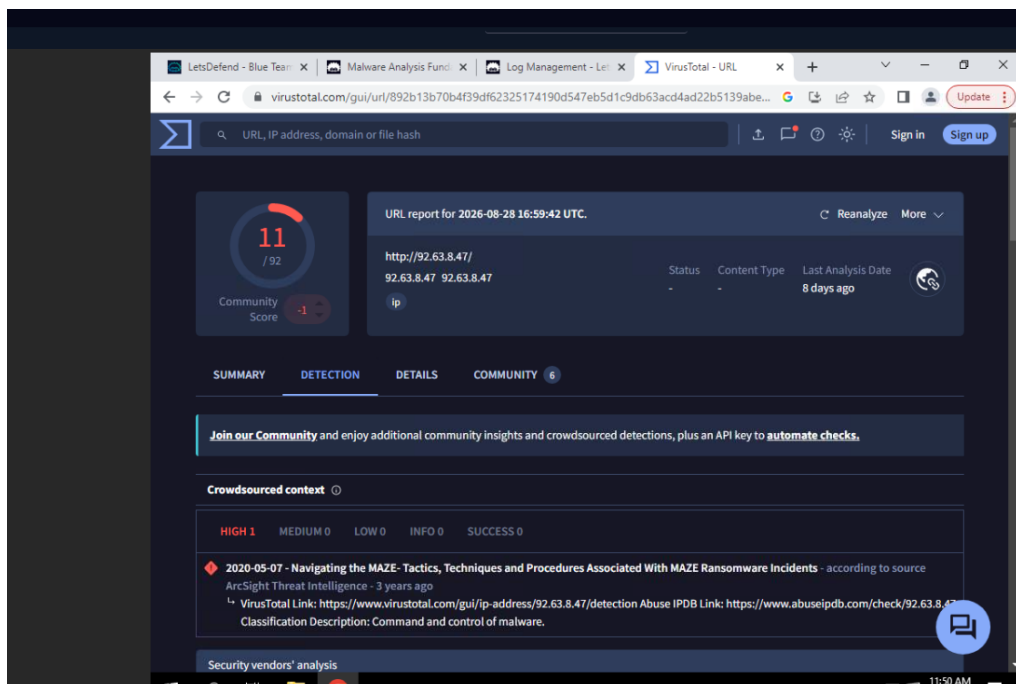


Figure 2 — VirusTotal detection result for the suspicious IP.

The community detection score (11/92) combined with a crowdsourced threat intel note (“Command and control of malware”) was enough to classify this as **malicious**.

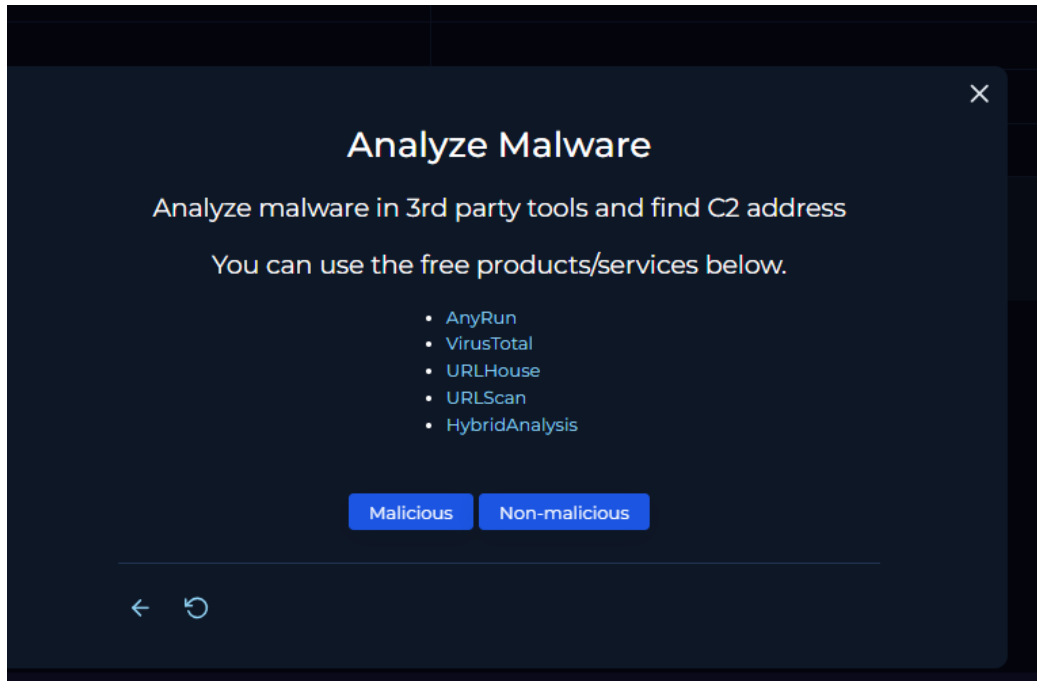


Figure 3 — Malware analysis tool selection and malicious verdict.

Step 3 — Check for C2 communication in logs

Searched Log Management for outbound traffic from **10.15.15.18** to the identified C2 address to determine whether the malware had actually “phoned home,” rather than assuming impact.

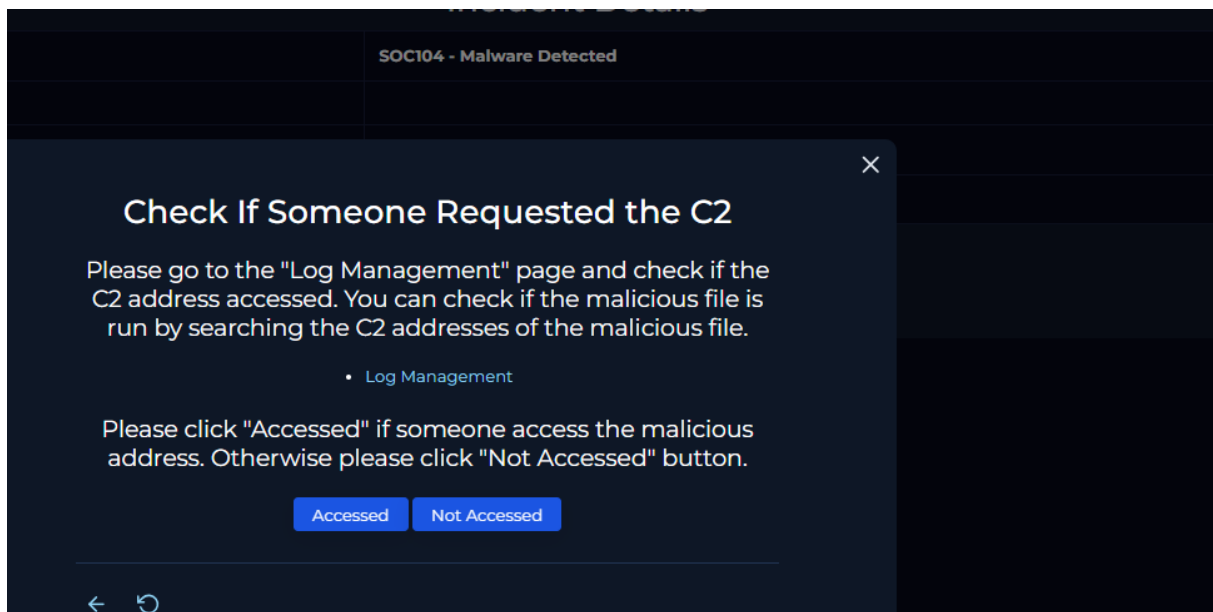


Figure 4 — Checking Log Management for C2 traffic from the infected host.

Step 4 — Contain the endpoint

Based on the confirmed malicious file and infrastructure, the affected host was isolated to stop further compromise or lateral movement while remediation continued.

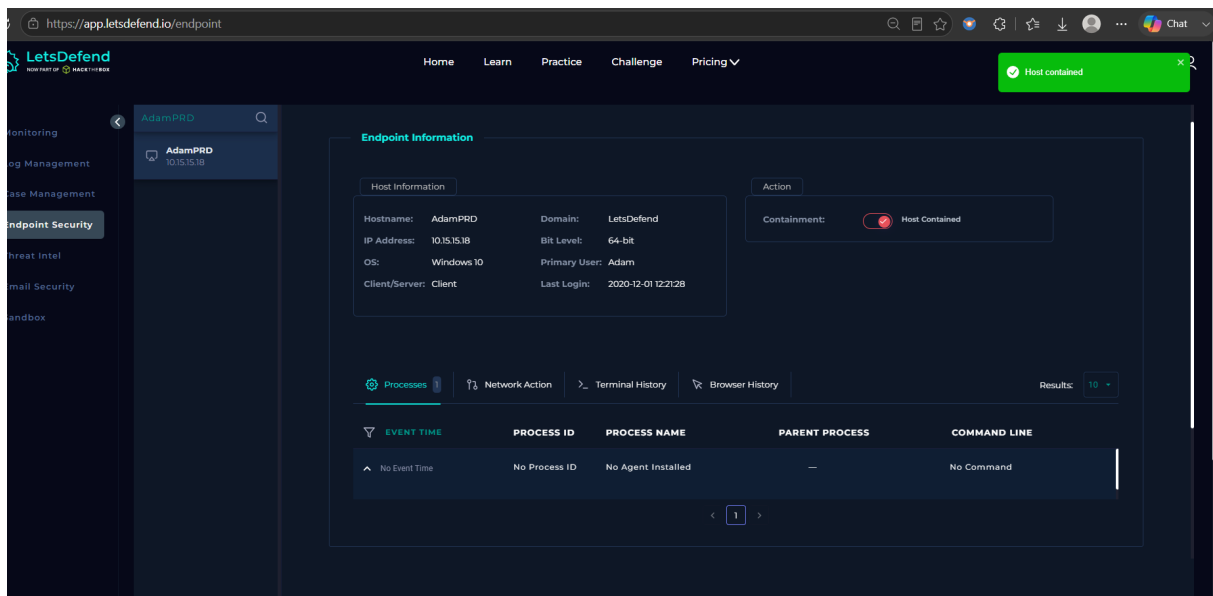


Figure 5 — AdamPRD endpoint isolated via containment.

Host Field	Value
Hostname	AdamPRD
IP Address	10.15.15.18
OS	Windows 10 (64-bit)
Primary User	Adam
Domain	LetsDefend
Containment Status	Contained

Step 5 — Close the case

With malware confirmed, malicious C2 infrastructure identified, and the host contained, the incident was submitted as a **True Positive**.

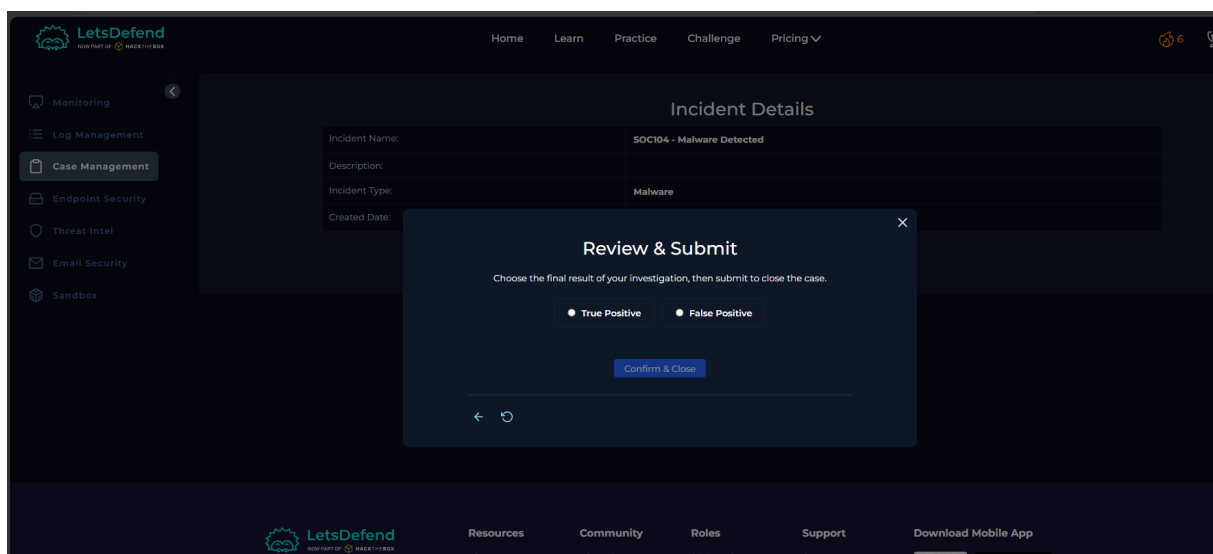


Figure 6 — Case closed as True Positive.

4. Investigation Performance Summary

LetsDefend scores each closed case against its official playbook. This case was closed with a perfect playbook match, confirming every required investigation step (triage, malware analysis, C2 lookup, containment, and closure) was completed correctly.

Metric	Result
Points Earned	+15
Playbook Success Rate	100%
Investigation Time	1h 10m

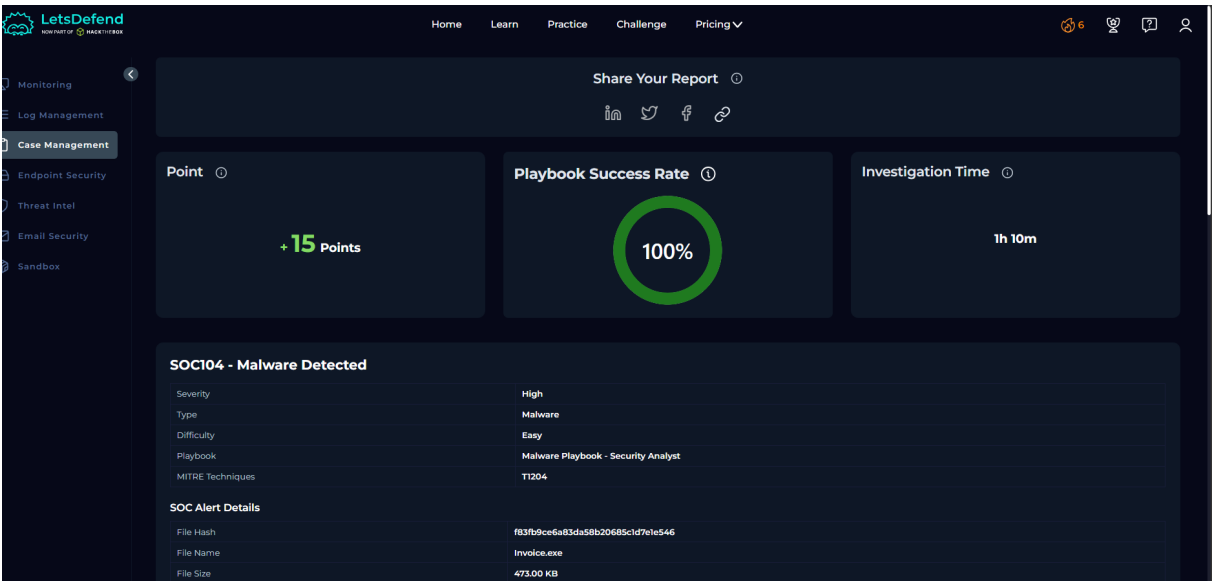


Figure 7 — Final case report: score, playbook success rate, and investigation time.

5. Indicators of Compromise (IOCs)

Type	Indicator
File Name	Invoice.exe
MD5 Hash	f83fb9ce6a83da58b20685c1d7e1e546
File Size	473.00 KB
Internal Host	10.15.15.18 (AdamPRD)
C2 IP Address	92.63.8.47

6. MITRE ATT&CK; Mapping

Tactic	Technique	Why it applies
Execution	T1204 – User Execution	The alert and file name (Invoice.exe) indicate the malware

relied on a user open

7. Findings & Recommendation

- **Verdict:** True Positive — a malicious executable ran on AdamPRD and network indicators link it to known C2 infrastructure.
- **Recommended remediation:** re-image or thoroughly scan AdamPRD, rotate credentials for the primary user (Adam), block 92.63.8.47 at the perimeter, and search the environment for the same file hash or destination IP on other hosts.
- **Lesson learned:** the file was initially allowed by the endpoint control rather than blocked — worth flagging for a detection-rule/AV signature review so similar lures are blocked automatically in future.

8. Skills Demonstrated

SIEM alert triage | Log analysis | Endpoint investigation | OSINT threat intelligence (VirusTotal) | IOC extraction | MITRE ATT&CK; mapping | Incident containment | Incident documentation

Disclaimer: This investigation was carried out in a simulated SOC environment (LetsDefend) for training purposes. All hosts, file hashes, and IP addresses belong to the training scenario and are not from a real organisation's environment.